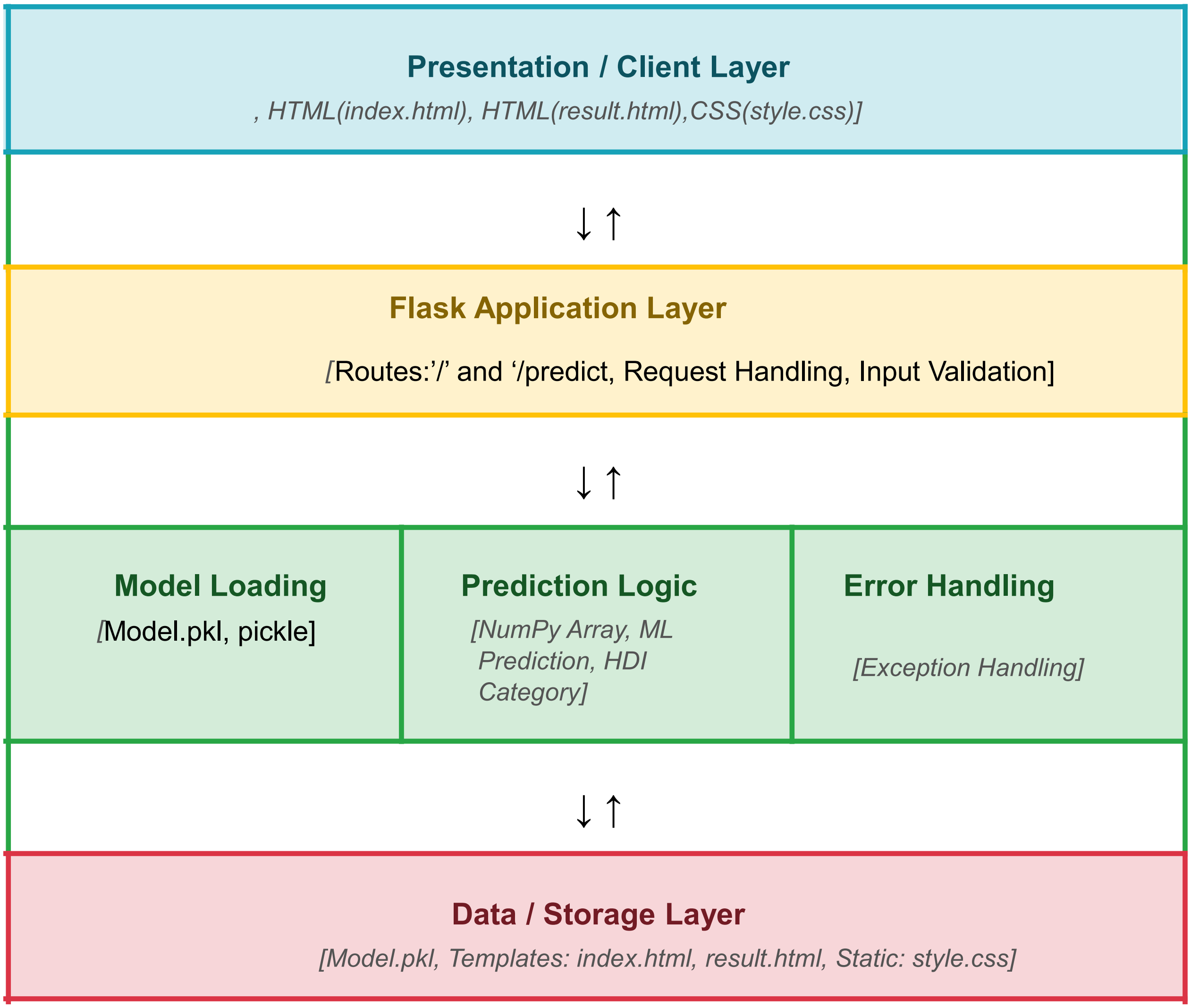
Solution Architecture

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Human Development Index (HDI) Prediction System follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a Flask-based web interface for user input and result display. The Application Layer processes the input and predicts the HDI score using a trained machine learning model, while the Data Layer stores the trained model (model.pkl) and the HDI dataset for efficient prediction.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter Life Expectancy, Mean Schooling, Expected Schooling, and GNI values. Displays predicted HDI score and category.	HTML5, CSS3, Flask Templates (Jinja2)
API Gateway	Receives HTTP requests from users, validates input, forwards data to the prediction model, and returns prediction results.	Flask (Python)

Core Logic Service	Processes user input, converts values into NumPy arrays, loads the trained ML model, predicts the HDI score, and classifies it into Very High, High, Medium, or Low Human Development.	Python, NumPy, Pickle, Scikit-learn
Database / Data Storage	Stores the trained machine learning model (model.pkl) and the dataset used during model training. Retrieves the model for prediction.	Pickle (.pkl), CSV Dataset